

TEMASEK POLYTECHNIC
SCHOOL OF INFORMATICS & IT
Diploma in Cybersecurity and Digital Forensics
AY 2022/2023 ASSIGNMENT REQUIREMENTS
(Oct Semester)

NETWORK SECURITY (CCF2C01)
SUBJECT LEVEL: 2

TIME ALLOWED: DUE ON WEEK 6

INSTRUCTIONS TO CANDIDATES

1. This paper consists of 13 pages (excluding the cover page).
2. This assignment constitutes 35% of the total marks for this subject.

Assignment Deliverables (35%, Total 100 Marks)

Your **individual assignment report** should include the following items (for both HQ and branch office, where necessary). Submit your **Packet Tracer implementation** together with your report.

Note: You may need to do some research on the Internet and/or do some self-study of the relevant lecture/lab materials to complete some of the assignment requirements. Remember to include **answers to the questions** in each part and **cite/acknowledge the sources of references** for your research.

- Part 1 –implementation issues **(5 Marks)**
- Part 2 – VLSM IP addressing design issues **(10 Marks)**
- Part 3 – DHCP, NAT/PAT and Wireless LAN implementation issues **(15 Marks)**
- Part 4 – Security implementation issues which includes the following **(Total 50 Marks)**
 - 4.1 Router Hardening **(15 Marks)**
 - 4.2 Switch Hardening **(10 Marks)**
 - 4.3 Firewall implementation issues **(Total 25 Marks)**
 - 4.3.1 IOS ZPF without DMZ implementation at BRANCH router **(10 Marks)**
 - 4.3.2 IOS ZPF with DMZ implementation at HQ FW router **(15 Marks)**
- Part 5 – Presentation Q&A **(Total 20 Marks)**
 - This activity will be done with the Project Presentation in week **7/8**

Do ensure you capture **ALL** the necessary details in your report and order them in the sequence of the tasks.

Assignment report (softcopy) and Packet Tracer implementation submission deadline via POLITEMail: Week 6 Monday (21 Nov) 9:00AM

Overview and Objectives

This assignment allows you to design and implement a secure internetwork by applying the internetworking and security concepts that you have learnt in NWS and other related subjects.

The assignment is based on a real-life scenario and it will also require you to do some independent research and implement feasible solutions based on the specific requirements of the scenario. It is crucial that you read and understand each requirement to ensure that the proposed solution and the accompanying report are complete.

This assignment requires that you accomplish the following:

- Plan the IP addressing scheme based on specific requirements (given by your lecturer). VLSM will be used to allocate IP addresses **efficiently**.
- Fill up the IP addresses table accordingly as in Table 1.
- Implement the various internetwork and network security requirements on the specific network devices (using Cisco Packet Tracer) and test to make sure that they work accordingly.
- Provide detailed documentation in a report as listed in the deliverables section.

This assignment is to be completed ***individually***. *Plagiarism will be dealt with severely according to school disciplinary guidelines*. The first page of your report **MUST** have the declaration as in Appendix A.

Note: A sample Packet Tracer file will be given to you with ***partial implementations*** that you can use to complete your implementation requirements as listed below. **You are required to use the assigned IP address given to you and make the necessary changes to the sample PT file.**

The network diagram with **SAMPLE** IP Address is as follows:

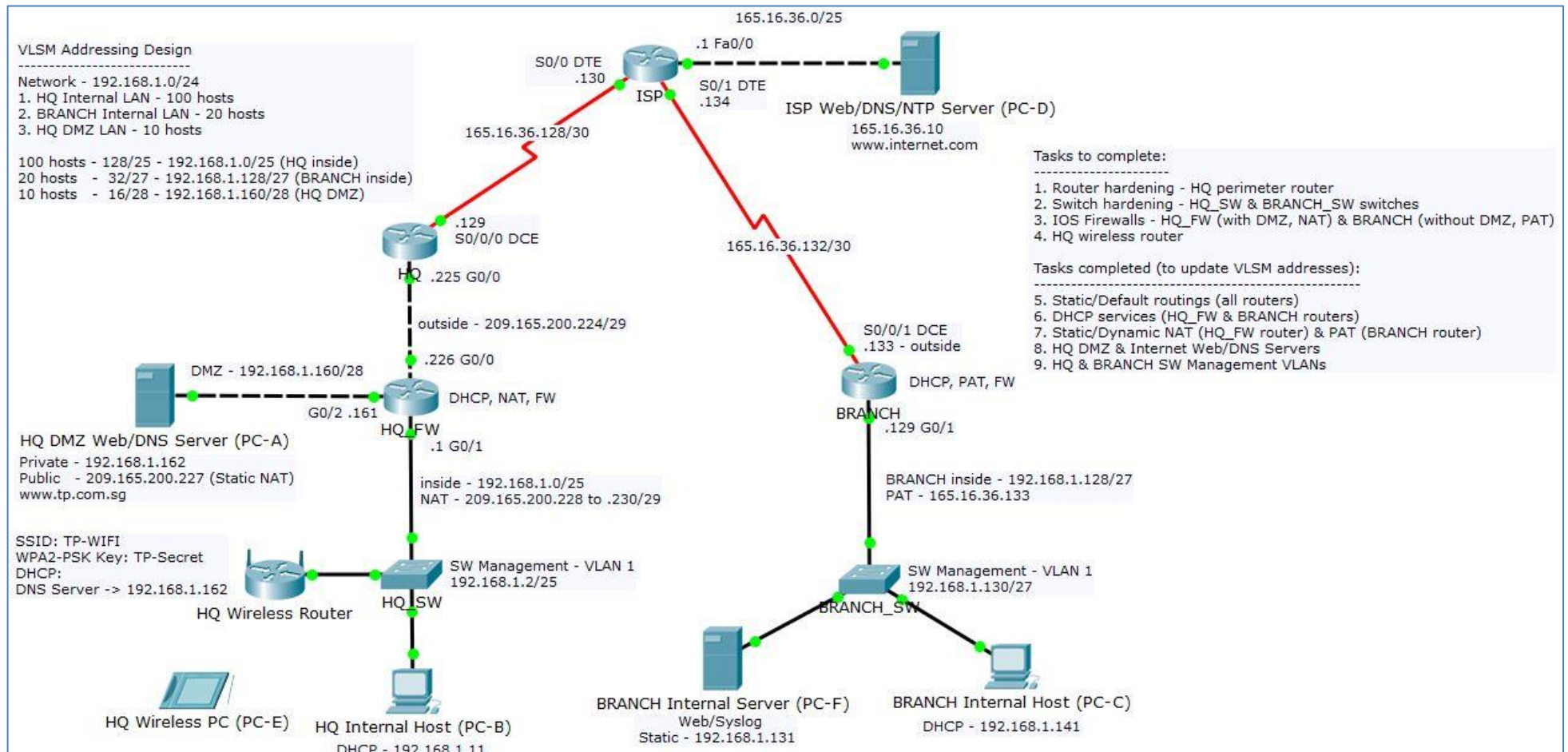


Figure A

Scenario

You have been tasked to design and implement a secure internetwork for a local company Tem Pora Pte Ltd (TP) with its main headquarter (HQ) located at **Shentonway** and a branch office (BRANCH) located at **Jurong Business Park**.

The company has strict security requirements for its corporate network. The chief network engineer has proposed all the important network components/equipments and a list of specific user requirements as follows Part 1 to Part 6 (refer to Figure A for further details).

Part 1: Static/Default routing implementation issues (5 Marks):

- 1 Internet connection is needed at HQ via a **1Gbps WAN connection** with public IP subnet address **165.16.36.128/30**.
- 2 Similarly the branch office has its own Internet connection via a **2.048Mbps WAN connection** with public IP subnet address **165.16.36.132/30**.
- 3 The company has been given the public network **209.165.200.224/29** to be used in the HQ. The HQ internal private networks will use Network Address Translation (NAT), whereas the branch office internal private networks will use Port Address Translation (PAT – mapped to its external public address **165.16.36.133/30**) to access the internet via the ISP router.
- 4 The company does not want to run any routing protocol (e.g. OSPF) with the ISP, so they require you to implement static/default routings between their perimeter routers and the ISP.

The static/default routing requirements have been implemented in the sample Packet Tracer file to satisfy the specific user requirements on routing issues. Identify the relevant *router IOS commands* in HQ, BRANCH and ISP routers and document them in your report to achieve the bandwidth and routing requirements as listed above.

Part 2: VLSM IP addressing design issues (10 Marks):

1. The company is given a pool of public IP addresses (legal Internet network addresses) **209.165.200.224/29** but are not sufficient to cater for all the hosts in the entire company network. Hence, you should assign private IP addresses to all the hosts in the company. The private IP address range (subnet) allocated and also the maximum hosts for each of the company LANs are as follows (assume ip subnet zero is enabled):

The private IP address range: _____

HQ private internal LAN: _____ hosts

HQ DMZ LAN: _____ hosts

Branch office private internal LAN: _____ hosts

To be
given
by your
lab tutor

Note: Your IP addressing design should start with the most number of hosts first.

2. Based on your IP addressing design, assign IP addresses to be used for each of the router/firewall interfaces accordingly (known entries from the network diagram have been filled up). Use Table 1 below to help you with the planning.

Table 1: Interfaces used and IP addresses

LAN / WAN	Router / Firewall	Interface	Network Address	IP address
HQ external public LAN (outside)	HQ perimeter router	G 0/0	209.165.200.224/29	209.165.200.225/29
	HQ firewall router	G 0/0		209.165.200.226/29
HQ DMZ LAN (Sacrificial LAN) (DMZ)	HQ firewall router	G 0/2		
	HQ DMZ Web/DNS server	Fa0 (NIC)		
HQ internal private LAN (inside)	HQ firewall router	G 0/1		
	HQ switch	Mgmt VLAN Interface VLAN1		
HQ wireless LAN	HQ wireless router	HQ wireless LAN connections	192.168.0.0/24	Obtain via DHCP service from HQ wireless router (start from 192.168.1.100)
Branch internal private LAN (inside)	Branch perimeter router	G 0/1		
	Branch switch	Mgmt VLAN Interface VLAN1		
	Branch internal Web.Syslog server	Fa0 (NIC)		
Public WAN between HQ and ISP	HQ perimeter router	S 0/0/0 (DCE)	165.16.36.128/30	165.16.36.129/30
	ISP router	S 0/0/0 (DTE)		165.16.36.130/30
Public WAN between Branch and ISP	Branch perimeter router	S 0/0/1 (DCE)	165.16.36.132/30	165.16.36.133/30
	ISP router	S 0/0/1 (DTE)		165.16.36.134/30
ISP Router LAN	ISP router	Fa 0/0	165.16.36.0/25	165.16.36.1/25

3. Assign the **second available IP address** from the respective subnets for HQ and BRANCH switch management VLAN interface (default to VLAN1).
Q: What is the disadvantage of this implementation, and how would you suggest to overcome it?
4. Assign the **third available IP address** from the BRANCH internal LAN as static IP address for the BRANCH internal Web/Syslog server. Similarly, assign the **second available IP address** from the HQ DMZ LAN as static IP address for the HQ DMZ Web/DNS server.
Q: Why do servers need static IP addresses?
5. Implement the above VLSM IP addressing scheme in the given sample **Packet Tracer file** by changing the IP addresses of the corresponding **router/switch interfaces** accordingly.
Q: List down any additional advantages/disadvantage of your proposed IP addressing solutions, if any. List down the unused/spare IP address range as well.

Part 3: DHCP, NAT/PAT and Wireless LAN implementation issues (15 Marks):

1. Implement **DHCP services** at both **HQ FW and BRANCH routers** so that users at the HQ and BRANCH internal LAN users can obtain their dynamic IP addresses accordingly based on your VLSM IP addressing scheme (refer to Table 1). Exclude the first 10 IP addresses from both HQ and BRANCH DHCP ranges

Q: What is purpose to exclude the first 10 IP addresses from both HQ and BRANCH DHCP ranges?

Also, HQ users should obtain their DNS service from **HQ DMZ Web/DNS server**, whereas BRANCH users should obtain their DNS service directly from the **ISP Web/DNS server**. Configure the respective DNS servers properly as well (**www.tp.com.sg** and **www.internet.com**).

2. Implement **static NAT** at **HQ FW router** to translate the HQ DMZ Web/DNS server's private IP address (as assigned in Table 1) to its public address **209.165.200.227/29**.
3. Implement **dynamic NAT** at **HQ FW router** to translate any HQ internal private LAN IP addresses to the public IP address range: **209.165.200.228 to 209.165.200.230/29**.
4. Implement **dynamic PAT** at **BRANCH router** to translate any BRANCH nternal private LAN IP addresses to BRANCH router's external public IP address **165.16.36.133/30**.
5. Implement wireless LAN with the following parameters to allow HQ wireless LAN users (**192.168.0.0/24**) to access the HQ corporate network resources:

SSID:	TP-WIFI
WPA2-PSK Key (Pass Phrase):	TP-Secret
Internet connection IP address:	Obtain from HQ DHCP server (HQ FW router)
Wireless LAN IP address:	192.168.0.0/24
	Obtain from wireless router DHCP service
	Start IP address: 192.168.0.100/24
	Range: 192.168.0.100 – 192.168.0.149 (max. 50)
	DNS server: HQ DMZ Web/DNS server

6. **IMPORTANT (Before you proceed further):** Test your implementations by showing that the **HQ wired and wireless users** (PC-B and PC-E) are able to access the HQ DMZ Web server (PC-A) as well as the ISP Web server (PC-D). Similarly for the **BRANCH wired users** (PC-C) as well. Also, **internet users** should be able to access HQ DMZ Web Server (via its public IP address – test with PC-D).

Identify the relevant *router IOS commands* in HQ, BRANCH and ISP routers and document them in your report to achieve the requirements as listed above. For wireless LAN implementation, provide the relevant *screenshots* with proper documentation. Also, provide relevant *screenshots* to show proper Web server access from the HQ wired/wireless PC as well as BRANCH wired PC.

Part 4: Security implementation issues (50 Marks):

The company wants to **harden their network devices** (routers and switches) as much as possible with the following proposed security solutions. They would also like to implement **IOS Firewalls** at HQ FW router (with DMZ LAN) and BRANCH router (without DMZ LAN).

4.1 Router Hardening (15 Marks):

1. Configure the following general router device security features at **HQ perimeter router**:

- Set the minimum password length of **10 characters**
 - Make sure all passwords are **encrypted**
 - Set the MOTD login banner to be "**Unauthorized access strictly prohibited!**"
 - Set enable secret password to be **cisco12345**
 - Create a user with username **Admin01** and the secret password **Admin01pa55**
 - Enable **AAA** and implement the following AAA services using local database:
 - Default login authentication method
 - Use local authentication as first option
 - Use enable password as backup option if errors occurs in local authentication
 - Configure the **console line** with the following parameters:
 - Privilege level 15 access
 - Log out after **15 minutes** of inactivity
 - Prevent console messages from interrupting command entry
 - Use the **aaa authentication default login method** for console login
 - Configure the **VTY (virtual terminal) lines 0-15** with the following parameters:
 - Privilege level 15 access
 - Log out after 15 minutes of inactivity
 - Prevent terminal messages from interrupting command entry
 - Allow remote access using SSH only
 - Use the **aaa authentication default login method** for vty login
 - Configure the router to **log login activities** with the following parameters:
 - Generate logging messages for all successful and failed login attempts
 - Configure enhanced login security by logging all failed attempts if the user experience **two failed login attempts within 30-second time span, and disable login for 1 minute.**
 - Configure the router as **SSH (Secure Shell) server** with the following parameters:
 - Domain name: **ccnasecurity.com**
 - Generate the RSA keys with **1024** as the number of modulus bits
 - Accept only **SSH version 2** connections
 - SSH timeout: **90 seconds**
 - Number of SSH authentication attempts: **2**
 - Verify **SSH connectivity** to HQ perimeter router from the **ISP Web/DNS server (PC-D)**
(Hint: Issue "**ssh -l Admin01 165.16.36.129**" from the command prompt window)
2. Repeat step 1 configurations above for both **HQ firewall router** and **BRANCH router**. Make sure you verify SSH connectivity from the **HQ or BRANCH internal host (PC-B or PC-C)**.
3. Configure a Synchronized Time Source using **NTP (Network Time Protocol)** as follows at **both HQ perimeter and BRANCH routers**: (ISP Web/DNS/NTP server will be the master NTP clock source for both routers):
- Turn on the **NTP service** at the NTP master with the following parameters:
 - NTP authentication key number **1** and password **NTPpassword**
 - Configure the following parameters at **NTP clients** (both HQ perimeter and BRANCH routers):
 - First noted that the current date/time at the NTP clients are set to **default** (Mar 1 1993) by issuing the "**show clock**" command
 - NTP authentication key number **1** with **MD5** hashing and password **NTPpassword**
 - Use the NTP trusted key above for authentication
 - Enable NTP authentication feature
 - Configure the router (as NTP client) to access its NTP master (**ISP Web/DNS/NTP server IP address**) with authentication key number **1**
 - Enable NTP client to **periodically update the calendar** with the NTP time source

- Verify the time on NTP client is now set correctly to the current date/time (issue the “**show clock**” command again)
4. Configure Syslog support at **BRANCH router** as follows:
 - Turn on **Syslog service** at the BRANCH internal server
 - Configure BRANCH router to log messages to the syslog server as follows:
 - First verify that BRANCH router can ping to its syslog server successfully
 - Enable the **timestamp service** (Hint: Use the “service timestamps log datetime msec” command)
 - Configure the syslog client (BRANCH router) to send syslog messages to its syslog server
 - Configure the logging severity level to **level 4 (warnings)** (Note: Packet Tracer may default to level 7 or debugging)
 - Verify that syslog messages are received at the syslog server
 5. **Wireless Router Hardening** - Configure **HQ wireless router** with the following parameters:
 - SSID: **TP-WIFI**
 - WPA2-PSK (Pass Phrase): **TP-Secret**
 - DNS Server: **192.168.1.162 (HQ DMZ Web/DNS server)**

Document **ALL** relevant *router IOS commands* used for hardening the HQ and BRANCH routers in the above steps, including all the relevant “*show*” *command outputs* in your report to achieve the router security requirements as listed above. Provide relevant *screenshots* where applicable, e.g. the syslog messages captured in the syslog server.

4.2 Switch Hardening (10 Marks):

1. Configure the following general switch device security features at both **HQ and BRANCH switches**:
 - Prevent HTTP access by disabling both the **HTTP server** and **HTTP secure server** (HTTP access to the switch is enabled by default) (Note: The “ip http server” and “ip http secure-server” may not be supported for switches in Packet Tracer)
 - Make sure all passwords are **encrypted**
 - Set the MOTD login banner to be “**Unauthorized access strictly prohibited!**”
 - Set enable secret password to be **cisco12345**
 - Create a user with username **Admin01** and the secret password **Admin01pa55**
 - Configure the **console line** with the following parameters:
 - Privilege level 15 access
 - Log out after 15 minutes of inactivity
 - Prevent console messages from interrupting command entry
 - Login with local user database
 - Configure the **VTY (virtual terminal) lines 0-15** with the following parameters:
 - Privilege level 15 access
 - Log out after 15 minutes of inactivity
 - Prevent terminal messages from interrupting command entry
 - Login with local user database
 - Allow remote access using SSH only
 - Configure the switch as **SSH (Secure Shell) server** with the following parameters:
 - Domain name: **ccnasecurity.com**
 - Generate the RSA keys with **1024** as the number of modulus bits
 - Accept only **SSH version 2** connections
 - SSH timeout: **90 seconds**

- Number of SSH authentication attempts: **2**
 - Verify **SSH connectivity** to HQ/BRANCH switches from the **HQ/BRANCH internal hosts (PC-B or PC-C)**
(Hint: Issue "`ssh -l Admin01 <SW-Mgmt-VLAN-IP>`" from the command prompt window)
2. Configure **switch port security** and **disable unused ports** at both **HQ and BRANCH switches** as follows:
- Disable trunking on all switch ports except G0/0 and G0/1 (i.e. set them to be **access ports**, and assume Gigabit Ethernet ports are reserved for **trunk ports**)
 - Enable **STP (Spanning Tree Protocol) PortFast mode** on all switch ports except G0/0 and G0/1 (**Q: What happen when a host PC is connected to a switch access port that is set to portfast mode, and why is it safe to configure portfast mode on all switch access ports?**)
 - Enable **BPDU (Bridge Protocol Data Unit) guard** on all switch access ports except G0/0 and G0/1 (**Q: What is the purpose of setting BPDU guard on switch access ports?**)
 - Apply basic **port security** on all switch access ports with the following parameters:
 - First enable switch port security
 - Set maximum MAC addresses to **2** and violation action to **shutdown** (default)
 - Use the **sticky option** to allow secure MAC address that is dynamically learned on a port to be saved to the switch running configuration
 - Disable all unused ports (i.e. shutdown all unused ports) in both switches (only **Fa0/18, Fa0/19 and Fa0/24** in **HQ Switch**, and **Fa0/5, Fa0/18 and Fa0/19** in **BRANCH Switch** should be in use)

(Q: Document ALL relevant switch IOS commands used for hardening the HQ and BRANCH switches in the above steps, including all the relevant “show” command outputs in your report to achieve the switch security requirements as listed above. Provide relevant screenshots where applicable, e.g. successful SSH login from the PCs).

4.3 IOS Firewall implementation issues (25 Marks):

The company wants to implement **IOS firewalls** at both HQ FW and BRANCH routers. HQ FW router should protect the internal LAN as well as having a **DMZ LAN** to house public servers such as Web / DNS / DHCP / mail servers, whereas BRANCH router will only protect its internal LAN.

4.3.1 IOS ZPF without DMZ implementation at BRANCH router (10 Marks):

Configure, apply and verify **ZPF (Zone-Based Policy Firewall)** configuration at the BRANCH router as follows. Note that there is no DMZ LAN in this router.

1. Create a ZPF policy that allows BRANCH internal LAN traffic to pass through this ZPF firewall to access external (public) networks as follows:
 - Create **INSIDE** and **OUTSIDE** security zones
 - Create an **inspect class-map** named **ALLOWED-PROTOCOLS** to match the traffic to be allowed from the INSIDE zone to the OUTSIDE zone. Because we trust the INSIDE zone, we allow ANY of the following main protocols (**tcp, udp and icmp**)

- Create an **inspect policy-map** named **INSIDE-TO-OUTSIDE**. Bind the **ALLOWED-PROTOCOLS** class-map created above to this policy-map. All packets matched by this class-map will be inspected
 - Create a **zone-pair** called **INSIDE-TO-OUTSIDE** that allows traffic initiated from the **BRANCH** internal LAN to the external (public) network. Apply the policy-map **INSIDE-TO-OUTSIDE** created above to this zone-pair
2. Similarly, another ZPF policy need to be configured to allow the return network traffic from external (public) network to return to its originated internal LAN as follows:
- First create an **extended access-list** that allow the return traffic from external public network to the original **BRANCH** internal LAN (i.e. source is **any** and destination is **BRANCH internal private LAN**) (**Q: Why destination is BRNACH private LAN addresses and NOT its equivalent public PAT address or 165.16.36.133?**)
 - Create another **inspect class-map** named **ALLOWED-TRAFFIC** to match both the traffic as well as the protocols from the **OUTSIDE** zone to the **INSIDE** zone (i.e. **match-all** condition that matches both the extended access-list and the **ALLOWED-PROTOCOLS** class-map configured in step 1 above)
 - Create another **inspect policy-map** named **OUTSIDE-TO-INSIDE**. Bind the **ALLOWED-TRAFFIC** class-map created above to this policy-map. All packets matched by this class-map will be inspected (**Q: What is the purpose of this policy-map?**)
 - Create another **zone-pair** called **OUTSIDE-TO-INSIDE** that allows the retrun traffic originally initiated from the **BRANCH** internal LAN to the external (public) network. Apply the policy-map **OUTSIDE-TO-INSIDE** created above to this zone-pair
3. Assign **BRANCH** router's **internal interface G0/1** to the **INSIDE** security zone and the **external interface S0/0/1** to the **OUTSIDE** security zone

(Q: Document ALL relevant *router IOS ZPF commands* used for ZPF implementation at the **BRANCH router in the above steps, including all the relevant “*show*” *command outputs* in your report to achieve the ZPF firewall security requirements as listed above. Verify the following conditions:**

- **BRANCH** internal LAN users (**PC-C**) can access both **ISP Web Server** and **HQ Web Server** (via its public address **209.165.200.227**)

Provide relevant *screenshots* where applicable, e.g. successful web access to **ISP and **HQ DMZ Web Servers**).**

4.3.1 IOS ZPF with DMZ implementation at HQ FW router (15 Marks):

Configure, apply and verify **ZPF** configuration at the **HQ FW** router as follows. Note that there is a **HQ DMZ LAN** to allow external public access to its Web server.

1. Create two ZPF policies that allows **HQ** internal LAN traffic to pass through this ZPF firewall to access external (public) networks, as well as allowing return traffic from external networks into the **HQ** internal LAN as follows:
 - First create three **INSIDE**, **OUTSIDE** and **DMZ** security zones
 - Create an **extended access-list** that allow the return traffic from external public network to the original **HQ** internal LAN (i.e. source is **any** and destination is **HQ internal private LAN**)
 - Create an **inspect class-map** named **ALLOWED-PROTOCOLS** to match the traffic to be allowed from the **INSIDE** zone to the **OUTSIDE** zone (and vice versa for return traffic). Because we trust the **INSIDE** zone, we allow **ANY** of the following main protocols (**tcp**, **udp** and **icmp**)

- Similarly, create another **inspect class-map** named **ALLOWED-INSIDE-TRAFFIC** to match both the return traffic as well as the protocols from the OUTSIDE zone to the INSIDE zone (i.e. **match-all** condition that matches both the extended access-list and the **ALLOWED-PROTOCOLS** class-map configured above)
 - Create an **inspect policy-map** named **INSIDE-TO-OUTSIDE**. Bind the ALLOWED-PROTOCOLS class-map created above to this policy-map. All packets matched by this class-map will be inspected
 - Similarly, create another **inspect policy-map** named **OUTSIDE-TO-INSIDE**. Bind the ALLOWED-INSIDE-TRAFFIC class-map created above to this policy-map. All packets matched by this class-map will be inspected
 - Create a **zone-pair** named **INSIDE-TO-OUTSIDE** that allows traffic initiated from the HQ internal LAN to the external (public) network. Apply the policy-map INSIDE-TO-OUTSIDE created above to this zone-pair
 - Similarly, create another **zone-pair** named **OUTSIDE-TO-INSIDE** that allows the return traffic originally initiated from the HQ internal LAN to the external (public) network. Apply the policy-map OUTSIDE-TO-INSIDE created above to this zone-pair
2. Repeat step 1 above for the following ZPF policies:
- Create a pair of ZPF policies that allow **HQ internal LAN users** to access the **HQ DMZ zone** (e.g. the HQ DMZ Web server) as follows:
 - Note that you have previously created an extended access list to allow return traffic (from anywhere) to **HQ internal LAN** in step 1 above
 - Create an inspect policy-map named **INSIDE-TO-DMZ** that bind to **ALLOWED-PROTOCOLS** (created in step 1) accordingly
 - Create another inspect policy-map named **DMZ-TO-INSIDE** that bind to **ALLOWED-INSIDE-TRAFFIC** (created in step 1) accordingly
 - Create a zone-pair named **INSIDE-TO-DMZ** and bind to the inspect-policy map INSIDE-TO-DMZ accordingly
 - Create another zone-pair named **DMZ-TO-INSIDE** and bind to the inspect-policy map DMZ-TO-INSIDE accordingly
 - Similarly, create another pair of ZPF policies to allow **external public users** to access the **HQ DMZ zone** as follows:
 - Create a new extended access-list to allow public traffic (from anywhere) to **HQ DMZ LAN**
 - Create a new **inspect class-map** named **ALLOWED-DMZ-TRAFFIC** to match both the public traffic (from anywhere) to HQ DMZ LAN as well as the protocols from the OUTSIDE zone to the DMZ (i.e. **match-all** condition that matches both the new extended access-list and the **ALLOWED-PROTOCOLS** class-map (created in step 1)
 - Create an inspect policy-map named **OUTSIDE-TO-DMZ** that bind to **ALLOWED-DMZ-TRAFFIC** (created in step 1) accordingly
 - Create another inspect policy-map named **DMZ-TO-OUTSIDE** that bind to **ALLOWED-PROTOCOLS** (created in step 1) accordingly
 - Create a zone-pair named **OUTSIDE-TO-DMZ** and bind to the inspect-policy map OUTSIDE-TO-DMZ accordingly
 - Create another zone-pair named **DMZ-TO-OUTSIDE** and bind to the inspect-policy map DMZ-TO-OUTSIDE accordingly
3. Lastly, assign HQ FW router's **interface G0/0** to the OUTSIDE security zone, **interface G0/1** to the INSIDE security zone, and **interface G0/2** to the DMZ security zone

(Q: Document ALL relevant *router IOS ZPF commands* used for ZPF implementation at the HQ FW router in the above steps, including all the relevant *“show” command outputs* in your report to achieve the ZPF firewall security requirements as listed above. Verify the following conditions:

- HQ internal LAN users (PC-B) can access both HQ DMZ and ISP Web Servers
- Public users (PC-D) and BRANCH internal LAN users (PC-C) can access HQ DMZ Web Server (via its public address 209.165.200.227)

Provide relevant *screenshots* where applicable).

Part 5: Q&A (20 Marks):

The Q&A session will be done during the Project presentation (in week 7/8). The purpose is to check your understanding of what you have done for the assignment.

----- END OF NWS ASSIGNMENT -----

Appendix A: The first page of the report must contain the following:

Network Security (CCF2C01)
Assignment Report Submission

Practical Class : P01 / P02 / P03 / P04 / P05 / P06

Submitted by: <Matric Number> <Full Name of student>

Date: dd /mm/yyyy

“By submitting this work, I am / we are declaring that I am / we are the originator(s) of this work and that all other original sources used in this work has been appropriately acknowledged.

I / We understand that plagiarism is the act of taking and using the whole or any part of another person’s work and presenting it as my/ our own without proper acknowledgement.

I / We also understand that plagiarism is an academic offence and that disciplinary action will be taken for plagiarism.”

NAME AND SIGNATURE OF STUDENT: